

Welcome to My

Project Documentation

A BRIEF REPORT OF MY PROJECTS

CREDITS / OWNER	NAME	YEAR	COLLEGE
	Manoj K	Final Year	Bangalore University

PROJECT 01

QR Shield

QR code safety screening with rule-based URL analysis

01 Problem Statement

QR codes can hide malicious URLs behind a simple scan. Users may open phishing pages, deceptive redirects, or unsafe destinations without seeing the link first.

02 Solution: QR Shield

Scans a QR code, extracts its destination, and checks it with rule-based analysis. It tells you whether the QR is safe or unsafe and only provides a navigation link when it is safe.

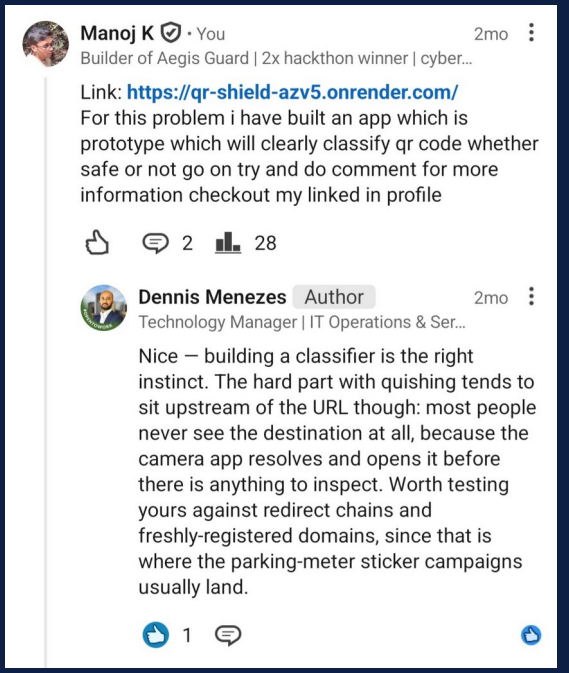
03 Mode of Operation

1. **Scan QR Code**
   User scans the QR code using the app.
2. **Extract Destination**
   The QR image is decoded and the URL is extracted for inspection.
3. **Rule-Based Analysis**
   The URL is checked against predefined rules, suspicious patterns and risk indicators.
4. **Decision & Result**
   Safe: show result and provide navigation link. Unsafe: show warning and never provide the link.

04 Tech Stack Used



05 Client Response



LIVE PROTOTYPE

https://qr-shield-azv5.onrender.com/

GITHUB REPOSITORY

https://github.com/manoj-24-24/QR-SHIELD-.git

PROJECT 02

Aegis Guard

Secure encrypted messaging with security analysis

01 Problem Statement

Current communication platforms do not have any built-in secure analysis of the messages they carry.

02 Solution: Aegis Guard

Aegis Guard first encrypts the message, then sends it for security analysis, so the recipient gets a proper, authenticated message.

03 Mode of Operation

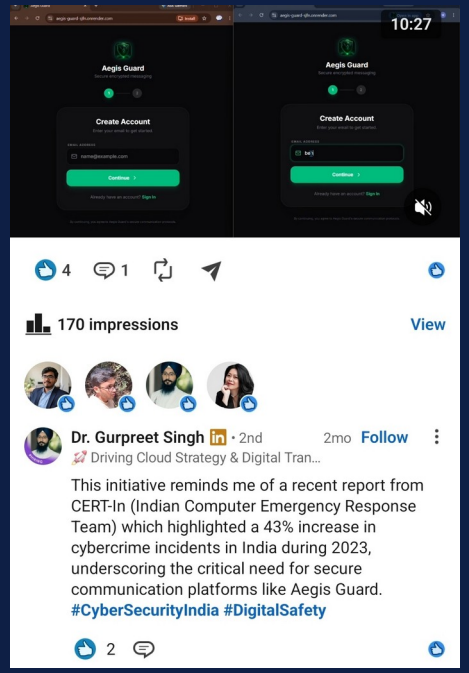
1. **Encrypt Message**
   The message is encrypted first, before anything else happens.
2. **Security Analysis**
   The encrypted message is sent for security analysis.
3. **Authenticated Delivery**
   Only a proper, authenticated message reaches the recipient.

04 Tech Stack Used



- TypeScript 97.4%
- HTML 1.4%
- Other 1.2%

05 Client Response



LIVE PROTOTYPE

https://aegis-guard-sjfn.onrender.com/

GITHUB REPOSITORY

https://github.com/krrish24-prog/Aegis-guard.git

PROJECT 03

Password Strength Tool

Website to check password strength and breaches

01 Problem Statement

Weak and reused passwords are an easy way into accounts, and most users cannot tell if their password is weak or has already leaked in a data breach.

02 Solution: Password Tool

A website that checks how strong a password is and compares it with a list of breached passwords, so users know right away if it is safe to use.

03 Mode of Operation

1. **Enter Password**
   User types a password into the website.
2. **Strength Check**
   The password is analysed to rate how strong it is.
3. **Breach Check**
   It is compared against the list of breached passwords.
4. **Show Result**
   Shows the strength rating and warns if the password is breached.

04 Tech Stack Used



JavaScript 56.2% CSS 25.2% HTML 18.6%

05 Key Features

PASSWORD STRENGTH



Weak Medium Strong

- ✓ **Strength check**
  Instantly rates how strong a password is.
- ✓ **Breached password list**
  Flags passwords that appear in known breaches.
- ✓ **Simple website**
  Open it in a browser and test a password.

LIVE WEBSITE

https://manoj-24-24.github.io/PASSWORD-STRENGTH-TOOL/

GITHUB REPOSITORY

https://github.com/manoj-24-24/PASSWORD-STRENGTH-TOOL.git